LAB 04 ANSWERS REPORT

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COMP3742, TONSLEY
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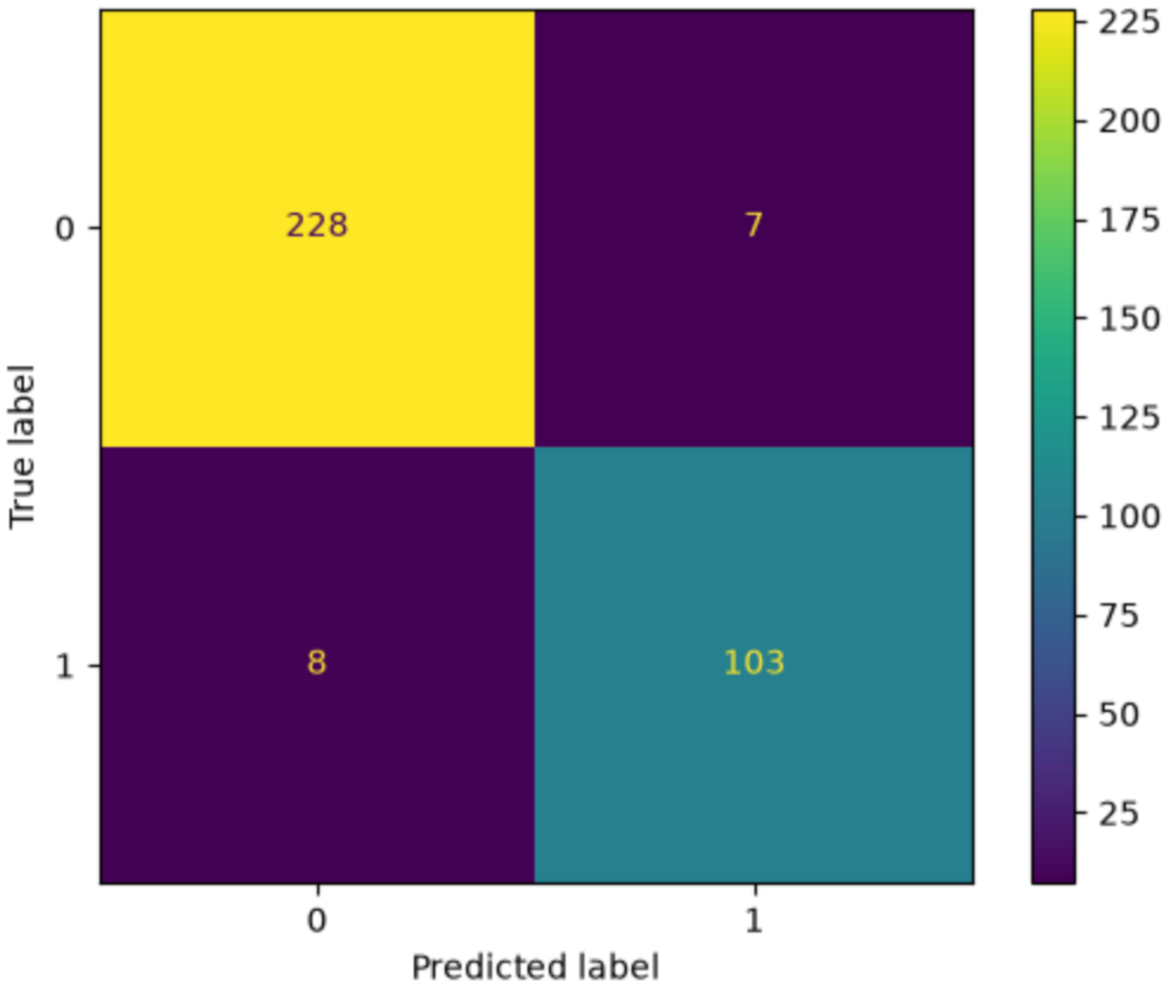
1. Implement Logistic Classification for binary classification using the Car dataset. Evaluate the model using accuracy and plot the confusion matrix.

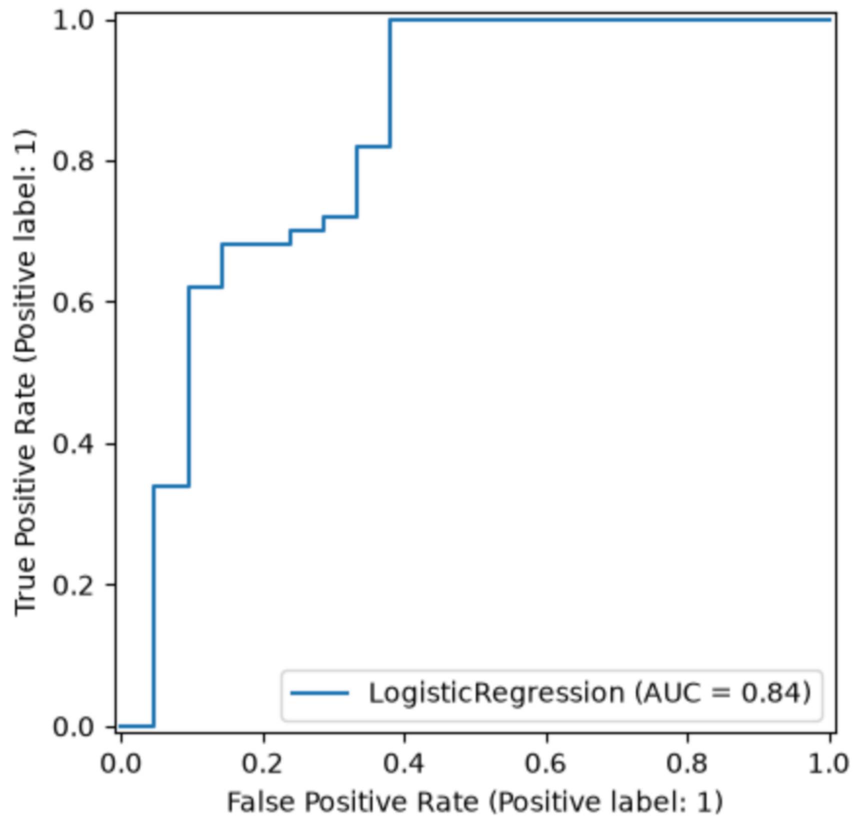


The model performed really well, correctly classifying 331 of 346 cars, with only 15 misclassifications, giving an accuracy of approx. 95.7%.

2. Implement Logistic Classification for multiclass classification using the Iris dataset. Use cross-validation to evaluate the model's performance.

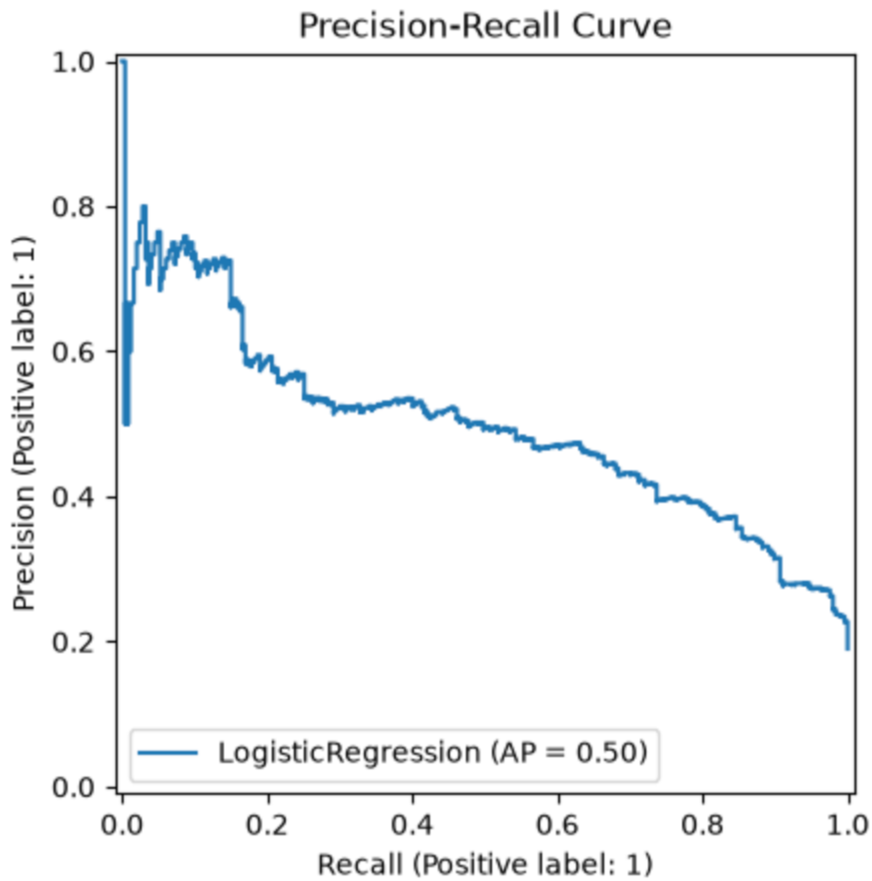
Using cross validation, the model scored an accuracy of 86.6%.

3. Implement Logistic Classification with L1 regularization on the Ionosphere dataset. Evaluate the model using ROC AUC score and plot the ROC curve.



The model illustrated solid classification performance, with a ROC AUC score of 0.836, with a relatively log loss of 0.412.

4. Implement Logistic Classification with class weights on an imbalanced dataset (e.g., the Wine Quality dataset). Evaluate the model using log loss and plot the precision-recall curve.



The model's precision-recall curve had an average precision of around 0.50. There is a clear trade-off between precision and recall, and the graph indicates there was moderate performance on the positive class.

5. Implement Logistic Classification with different solvers (liblinear, lbfgs, saga) on the MNIST dataset. Compare the performance of each solver using cross-validation and visualize the results